Analysis of Portfolio Risk Across NIFTY Sectors During Macroeconomic Stress Events

YASHPAL GUPTA S
Dept. of Computer Science Engineering
Dayananda Sagar University, Bengaluru, India
yashpalgupta-cse@dsu.edu.in

Akshata I Kallimath
Dept. of Computer Science Engineering
Dayananda Sagar University, Bengaluru, India
eng23cs0016@dsu.edu.in

Basanagouda
Dept. of Computer Science Engineering
Dayananda Sagar University, Bengaluru, India
eng23cs0028@dsu.edu.in

Guru Kumar
Dept. of Computer Science Engineering
Dayananda Sagar University, Bengaluru, India
eng23cs0070@dsu.edu.in

Indushree D N
Dept. of Computer Science Engineering
Dayananda Sagar University, Bengaluru, India
eng23cs0080@dsu.edu.in

***Abstract*—Macroeconomic events, geopolitical tensions, and monetary policy decisions cause volatility and alter risk levels in market segments. Some previous studies have explored sectoral volatility, spillovers, portfolio optimization and stress testing in the context of the Indian stock market during COVID-19 and the Russia-Ukraine conflict, as well as with changes in monetary policy. While such studies are available, they lack an integrated benchmark that combines sector level information with multiple stress events. There is a need for studies that provide benchmarks on sectoral risk levels, portfolio drawdown metrics and standardized risk labels. This study introduces NiftyStress-44, which is a labeled dataset designed to help understand risk profiles across the 11 sectors of the National Stock Exchange (NSE) over the course of 4 different macroeconomic events during the period 2020-2023. These events are the COVID-19 crash, Russia-Ukraine conflict, the Reserve Bank of India's rate hiking cycle, and the U.S. Federal Reserve's monetary policy tightening. The data will be sourced using public financial databases to compute maximum drawdown, Value at Risk (VaR) at the 95**

***Index Terms*—Distributed Machine Learning, Collaborative Learning, Federated Learning, Privacy Preservation, Non-IID Data, Data Heterogeneity, Model Convergence, Decentralized Systems, Fake Certificate Detection**

1. INTRODUCTION

Economic growth is closely linked to how efficient and functional a countrys financial systemsre. Financial systems are very important for investment for deciding where to put resources and for the process of capital formation. In today's economic environment the ability of financial systems to do these jobs is very sensitive to shocks. These shocks can come from economic factors or even from geopolitical conflicts. Public health emergencies and even the way that money supply is regulated can cause market volatility. Market volatility can shift investor sentiment in a direction. Market volatility shift can expose portfolios to financial losses. Different industries feel these shocks in ways and to different degrees. Studying these industry differences can help explain why some parts of the economy are more vulnerable. Knowledge of industry differences can help design risk management strategies for portfolios. The National Stock Exchange (NSE) and its Nifty Indices are sources of information about industry differences, in India.

In 1952, Markowitz presented Modern Portfolio Theory to help guide investors on how to construct an optimal investment portfolio while considering trade-offs between risk and expected return. While there are benefits to diversifying your investments, there are situations in which this method falls short. In times of market volatility, correlation among assets changes. Because of this, it is necessary to consider the duration and potential loss of your investments as well as the expected market return when thinking about investment portfolio construction. It's also crucial to understand the implications of cross-border financial transactions and investments when looking at emerging markets.

The COVID-19 pandemic has been one of the important recent cases of a global shock to the financial markets. The outbreak and the subsequent lockdowns disrupted activity and supply chains, as well as employment and consumption patterns. The Indian market saw a fall in the value of shares in the February-April 2020 period. Despite the liquidity measures and the lowering of the repo rate by the Reserve Bank of India (RBI) there were effects on different sectors. Technology-intensive companies were not in the position as far as demand was concerned as compared to more traditional heavy industries. Moving consumer goods and other relatively defensive sectors differed in their response. All this heterogeneity suggests that examining the behavior of aggregated indices is not enough when it comes to understanding risk exposure. It is important to look at the characteristics of different industries to understand how their portfolios were affected by the pandemic

and what the risks were.

The situation with the Russia-Ukraine conflict, which began in February 2022 with Russia launching an invasion of Ukraine is another case of a shock that the financial markets have recently experienced. It has implications for energy prices, commodities, international trade and capital flows. For the market the fluctuation in the global prices of commodities and the impact on capital inflows affect different sectors unevenly. For example the energy and metal sectors may benefit from prices but other areas of industry may suffer from increased costs and a decrease in values. On the hand the situation is complicated by the fact that different industries have their particular exposure to geopolitical risks. Therefore it is important to examine the February-April 2022 period to identify differences, between the sectors.

Monetary policy developments pose a financial stress. The central bank has increased its repo rate during the period of May 2022 to February 2023. The policy rate in the central bank rose from 4.0

The fourth event is that of the US Federal Reserves tightening cycle that ran from March 2022 to December 2023. The changes in US interest rates can result in effects on money flows, the way money circulates globally, the risk appetite and the value of the currencies. Countries like India can experience the impact via the changes in the inflows of investors and the attractiveness of financial assets abroad. These changes can cause pressure on the stock prices. Lead to different responses in Indian sectors. For example, information technology companies may respond to changes in the demand, the currency values and how they are valued. Other sectors may be affected by the domestic issues or the prices of goods. Despite the US Federal Reserves tightening period occurring at the same time as the conflict in Russia and Ukraine and the RBI rate increase, looking at it as a separate event allows a clear way to compare how different sectors were affected in different economic situations.

Researchers have studied stock markets' volatility and how the volatility spreads around the world. They have explored how sectors are consistent over time, how investment decisions can be made and how to test for financial stress. They have made use of methods such as ARCH/GARCH, vector autoregression and detrended fluctuation analysis to understand how markets behave during uncertain times. Other studies have looked at how investment portfolios can be built by considering risk. They have made use of ratios, like Sharpe, Sortino and Calmar. New methods that make use of machine learning have provided ways to model complicated relationships and create scenarios that indicate how the financial system could act in bad situations. However, there is a need to create a dataset that looks into how portfolios lose value during different economic stress periods and how different sectors behave in the Indian situation. Putting together the risk indicators into one system allows better comparisons that are not easily found when looking at work focused on a single crisis or a single way of measuring volatility.

To meet this research need this study will build NiftyStress-44. NiftyStress-44 is a portfolio risk dataset that looks at risk for each sector. It covers eleven NSE sectors and four macroeconomic stress events during the years 2020 to 2023. To create this dataset the study will pull market data from 2019 to 2024. The data will be downloaded from financial databases and processed using Python. In detail the study will report four indicators of portfolio downside risk: maximum drawdown (MDD) Value at Risk (VaR) at a 95 percent confidence level, recovery days and the annualised change in the Sharpe ratio compared to the previous period. MDD is calculated to find the drop from a peak to a trough between consecutive closing values of the portfolio. VaR is estimated for the five percent of the daily return distribution. Recovery days count how many days are needed for the portfolio to return to its peak value after an event. The change in the Sharpe ratio measures the return per unit of risk during the stress event. Each sector-event observation will be classified as medium or low risk based on MDD thresholds.

By creating this set of forty-four sector-level observations the study will meet the need to compare sector-specific vulnerabilities and financial resilience across the four macroeconomic stress windows.

To do this the random forest classification algorithm will be used as the baseline approach. The random forest classification algorithm will rank each sector-event observation by its predicted risk level. The study will evaluate the performance of the forest classification algorithm using validation procedures and relevant performance metrics.

Overall this research project will help analyse stress compare portfolio risk and explore machine learning methods in emerging market finance. In particular the study will improve understanding of portfolio risk by combining several risk factors into one dataset that will serve as the basis, for further quantitative analysis.

2. LITERATURE SURVEY

Financial markets can experience disruptions due to situations, political events, decisions on money supply and unexpected global happenings. In a stressed scenario, changes in the perception of investors, changes in prices, ease of sale and the co-movement of assets can have a significant effect on a diversification of investments portfolio performance. Although historical data on stock prices can provide information on price changes, analyzing returns may not provide information on losses during very stressful scenarios. Because of this, experts have started examining how different sectors of the market behave, how volatility is transmitted between areas, measuring risk of losses and using data to analyze financial stress.

Studies that are related to this research can be divided into four groups: how different parts of the market act during bad times, measuring the risks in a collection of investments using computer learning to test for financial stress and how the economy affects the stock market. These studies can help understand how different parts of the market respond to difficult situations. Each study has different goals, different data, different ways to measure things and different ways to

check how good the results are. This study uses what has been learned from these studies to analyze the risk of losses in Indian market sectors during four different times of economic stress.

A. Sectoral Market Behaviour During Financial Crises

Research on sectoral market behaviour shows that it is useful to consider each industry on its own rather than considering the entire stock market. Different sectors have ways to make money different costs different financing needs and different responses to changes in the economy. Therefore a common shock to the market can have different effects on different sectors.

Studies that analysed the COVID-19 pandemic investigated how the Indian stock market behaved during a public-health and economic crisis that had never happened before [1 2]. The pandemic altered how businesses worked how people purchased things how goods were transported how factories ran and what investors expected. Rules restricting movement and business activity put pressure on industries that are highly sensitive to the economy. At the same time other sectors witnessed shifts in demand for health services in the use of digital tools and in the purchase of essential goods. These differences reflect the fact that an index that covers the market cannot fully capture the risks each sector faces.

Sectoral studies of the COVID-19 period also highlight the importance of examining volatility and the dispersion of returns. Higher volatility indicates uncertainty but that it does not disclose how big the biggest loss for an investor can be. One sector can experience ups and downs without experiencing the largest drop from their peak to their lowest point while another sector witnesses a long decline that takes a long time to recover. This demonstrates that it is useful to add measures of risk for a portfolio in addition to the usual return and volatility statistics.

Other studies investigate how markets are interconnected and how volatility is transmitted between sectors and other assets using methods that allow for statistics to change over time such as TVP-VAR approaches [3]. These approaches allow researchers to uncover how the relationships between variables change over time and whether a shock in one market or sector affects other markets.

Research that uses models that analyse variables simultaneously such as VAR-GARCH-BEKK also examines how volatility is passed from one market to another and how international markets interact with one another [4].

These approaches provide insights into how financial risk's dynamic and how markets are related.

While studies of interconnectedness and spillovers are useful for identifying how risk is passed from one market to another they usually aim at a question than studies that label risk for a portfolio during a specific event. Knowing which sectors transmit volatility does not necessarily reveal which sectors suffer the losses or which ones take the longest time to recover.

The results, from these studies therefore support examining each sector on its while also allowing for comparisons of downside risk consistently across many stressful events.

B. Geopolitical and Macroeconomic Stress Events

In addition to public-health crises, geopolitical conflicts and changes in money policies can also affect how stock markets perform. Studies looking at the Russia–Ukraine conflict checked how it connected with sectoral indices using return analysis, basic statistics and comparing numbers with methods like paired t-tests [5]. The conflict created uncertainty about energy prices, supplies of goods trade around the world and the overall economy. These things can affect industries in different ways depending on how much they rely on imported materials, energy costs, outside demand and what people buy at home.

The value of this research for the study is that it looks at a specific event. A geopolitical problem is not the same as a pandemic. It has ways it spreads and affects different areas. So looking at different stress events could help understand which industries are more at risk than looking at just one crisis. However ways of defining events different times to watch and different ways to measure performance can make it hard to compare.

Money policy is another reason for stress in financial markets. Changes in interest rates can affect how much it costs to borrow how money is valued, investment choices and what people think about growth. Research looking at how the Reserve Bank of Indias repo rate changes connect with the Nifty 50 used things like correlation and covariance to study money policy and how stock markets move [6]. These kinds of studies help show how the economy and stock markets are connected.

In the way when the United States Federal Reserve raises rates it can affect the whole worlds financial situation through expectations about interest rates, where money goes, how currencies change and how willing investors are to take risks. For a country like India, which is still growing these things can affect how much money is invested and how much is spent locally.

The current research looks at the crash caused by the COVID-19 pandemic the shock from the Russia–Ukraine conflict the RBI rate-hike period and the time when the US Federal Reserve increased rates. These are four times of economic stress. They each come from sources of uncertainty and are the basis for comparing how different industries fell during different times.. Because some of these times overlapped, especially in 2022 and 2023 it is not always clear which event caused the losses. So the study should be seen as comparing times of events. Not, as proving that each event caused the results.

C. Portfolio Risk Measurement and Downside Risk Analysis

Portfolio risk assessment considers performance characteristics beyond those related to returns. The simplest performance indicators identify the degree of dispersion of returns, for example, the standard deviation. Risk – adjusted performance

indicators relate the return to the amount of risk, for example, Sharpe's ratio, Sortino's ratio, and the Calmar ratio. These indicators allow comparison of the performance of different portfolio strategies, each of which has its own risk-return relationship.

The Sharpe ratio shows the relationship between the standard deviation and the excess return, Sortino's ratio shows the same relationship but for the downswing. In turn, the Calmar ratio reflects the relationship between the return and the maximum drawdown. Although each of these three indicators of performance is useful in comparing the risk and return of different strategies, it seems that the risk of a portfolio is a more complex concept than these indicators usually express. Although the three indicators above relate risk to return, each of them reflects only one source of risk, and therefore fails to fully reflect the relationship between, for example, standard deviation and total return, or downswing, or maximum drawdown.

One of the simplest indicators of portfolio risk is the maximum drawdown (MDD). The MDD reflects the distance from the highest value to the lowest value that the portfolio can experience during a specified period. Since the MDD reflects the decline relative to the previous peak, this indicator allows assessing the risk associated with the drop in the value of the portfolio. Another indicator of portfolio risk is Value at Risk (VaR), which reflects the loss that would occur with a given probability during the specified period. The most common choice is the daily VaR at the 95

It seems that both the MDD and the VaR are useful tools for assessing the risk of a portfolio. However, using these indicators, one cannot answer the question of how long the decline in the value of the portfolio will last. Therefore, another indicator of portfolio risk, which reflects the amount of time between the peak and the lowest point after that, is the recovery period. The recovery period, like the MDD, is a measure of the duration of a portfolio decline. While holding two portfolios with the same MDD, one will have to wait for the recovery for a much longer time than for the other. Thus, the length of the period in which the value of the portfolio decreases is also a critical risk indicator because it affects the ability of the investor to withstand this decline.

Therefore, in order to fully analyze the risk of a portfolio, it is necessary to consider several indicators at once, reflecting different aspects of portfolio risk. The four indicators of portfolio risk discussed above, namely the maximum drawdown, VaR, recovery period, and Sharpe ratio, allow for a more complete assessment of the risk profile of the portfolio.

D. Machine Learning and Financial Stress Testing

Machine learning has been used more and more for identifying risk predicting it and testing stress. While normal statistical models can still be used to examine relationships and measure risk machine learning algorithms are better at finding connections and doing classification tasks.

The existing literature on machine learning for stress-testing includes studies that look at dimensionality reduction and representation-learning techniques such as PCA, autoencoders, VAE and risk metrics like VaR and Expected Shortfall [8]. The two latter concepts are risk metrics with Expected Shortfall defined as the average loss beyond a VaR threshold, which is set by a percentile or confidence level [8]. Thus it is possible to use both risk metrics to fully describe financial risk exposure. However the effectiveness of such an approach depends on the quality of the data set the choice of stress-testing scenarios and the evaluation design.

Meanwhile the literature on machine learning for financial risk identification includes work that looks at classification algorithms, like Random Forests [9]. Random Forest uses decision trees together to estimate the link between input features and a target variable which makes Random Forest good at finding non-linear relationships. In this study the author suggests using a Random Forest classifier to see if the chosen portfolio-risk indicators can classify sector-event observations by risk level.

However when using machine learning for financial risk identification one must remember that the nature of instruments and market forces can shift a lot during severe market stress making old statistical patterns and rules of thumb useless. Therefore it is essential that the classifier was trained on data that include a financial crisis period if that scenario matters for the study.

Moreover the performance of the suggested classifier can be measured with metrics, such as precision, recall and F1-score besides accuracy. At the time the suggested approach should be seen as a proof-of-concept only because the data set used in this paper is small (44 sector-event observations or 11 sectors times 4 event categories). Thus using Random Forest for classification is suitable, for exploratory analysis. Also the choice of features must be careful to avoid target leakage. For example if MDD is the target variable it should not be part of the feature set because then the Random Forest would always identify it as the target.

E. Existing Market Datasets and Data Availability

Historical market OHLCV data are an input for quantitative market analysis. Available data sets and financial data services enable the reconstruction of daily price histories for individual securities and market indices alike. These data help compute returns volatility figures, portfolio performance and other risk-oriented indicators.

Nevertheless the availability of pricing data does not automatically mean a pre-processed crisis-risk data set is present. Identifying stress windows, sector-specific constituents or representative instruments calculating coherent portfolio returns estimating downside-risk figures and assigning risk categories may require separate research efforts. Because of these variations the elements of a crisis-risk data set can differ a lot across studies making comparison difficult.

In the research project the question is not only about whether historical price data of the National Stock Exchange (NSE) exist. The project seeks an assessment of whether

a standardized set of Indian sector portfolios a clearly defined set of macroeconomic stress windows a set of comparable downside-risk measures and sector-event specific risk labels can be found in publicly accessible data repositories such as Kaggle or Mendeley Data. The absence of such a pre-processed data set should be shown by a search of relevant data repositories and a review of their descriptive documentation not by guessing from a few papers. Similarly statements, about the lack of MDD, VaR and recovery-time measures should be based on the extent of the literature reviewed.

F. Research Gap and Proposed Contribution

The research that was reviewed gives information about how different parts of the market behave how markets are connected what happens when there are political problems what happens with money rules how to make the best set of investments and how to use computer learning to check how well investments can handle tough times.. The studies that were mentioned before look at these topics from different ways of thinking. Some look at how marketsre connected and how changes spread others look at specific events or how different parts of the market do and others work on how to make the best set of investments or how to build models for risk.

Four areas are the reason why this study is being done.

First the old data sets that have open high low, close and volume numbers mostly give information about prices. How much is bought and sold. A researcher might still need to make their own observations about events and calculate risk numbers for portfolios. How much these data sets already have this full ability needs to be checked by looking through a collection of data.

Second how different parts of the market react could be different when there are kinds of tough times. This makes it helpful to have a way to compare risk in situations. Even if a study looks at time periods using the same definitions it can't show what causes the risk when there are overlapping big economic problems without more work.

Third the current research is trying to make a table that shows the risk level for each event in market sectors. This table would allow comparing risk levels across sectors and events using the same way of grouping instead of looking at each event with different ways of measuring.

Finally the study brings together MDD, 95

To fix these areas the NiftyStress-44 data set that is being proposed will have 44 examples of sectors and events. These examples cover 11 sectors from the National Stock Exchange across four times when the economy was under stress. Old price data will be used to calculate returns and make sure that the data for each sector is the same. MDD will be the base for giving High, Medium and Low risk labels while VaR how long it takes to get to normal and how the Sharpe ratio changes will show more details about the risk.

The main idea of this study is making a event-based data set and creating a basic way to look at risk. By using the steps, for all sectors and times of stress the study wants to make it possible to compare how much risk there is in the Indian stock market. The results of the study will be limited by the sectors chosen how events are defined how much data is available and the small number of events that have been labeled.

G. Summary

The Literature Shows That Volatility From Financial, Geopolitical and Monetary Policy Sources Varies Across Sectors: A Shared Dataset for Volatility Analysis Stress Testing and Drawdown Research Thesis

The literature proves that financial crises, geopolitical conflicts and changes in policy can affect sectors in different ways. The literature shows that these events create levels of volatility for each sector. Existing methods for volatility analysis portfolio optimization and machine-learning stress testing give tools yet they focus on separate parts of financial risk.

The present research brings these ideas together by creating a shared sector-event dataset that includes severity, VaR, recovery time and changes, in risk-adjusted performance. This framework is designed to make it easier to compare sector volatility during macroeconomic stress events making the comparison more systematic, clear and repeatable.

"'latex

3. Proposed Methodology

The aim of the proposed research is to build NiftyStress-44 a labeled dataset that helps examine the risk of portfolio drawdown for eleven NSE sector portfolios during four stress events in India. The research method includes gathering market data creating sector portfolios computing financial risks labeling each risk and using basic machine learning classification.

The main goal of the research is to create an repeatable framework that measures the risk of each sector when markets are stressed. The labeled dataset that the research builds is not like datasets; it not only shows how the selected stocks move over time but also records the Maximum Drawdown (MDD) the Value at Risk (VaR) at 95 percent confidence the recovery time and how the Sharpe ratio changes. Using the MDD limits the research assigns every sector-event observation, to one of three risk categories: High, Medium or Low.

The major contributions of the proposed methodology are as follows:

- Creation of the sector-event data set encompassing 11 NSE sectors and 4 macroeconomic stress events with 44 sector-event realizations in total.
- Gathering and preprocessing of the daily OHLCV data for the 2019–2024 period using the data available in the public domain.
- Calculation of 4 risk indicators including Maximum Drawdown, historical Value at Risk (95%), recovery days, and Sharpe ratio change.
- Design of the 3-class risk-labeling scheme based on the selected Maximum Drawdown thresholds.

- Validation of the risk indicators via quality-control procedures, manual checking, and comparison to the market realities during the stress periods of interest.
- Development of the baseline Random Forest classifier to evaluate the association between the sector-event attributes and risk labels.
- Compilation of an archival data set, data dictionary, processing instructions, and risk-label matrix for subsequent use in financial stress analysis by other researchers.

A. System Overview

The proposed system is built around a series of steps that take observations and turn them into a final dataset labeled with portfolio risk information. This process happens in six stages: identifying stress windows gathering market data, preprocessing the data calculating risk metrics and labels and running a baseline model.

To begin the research defines four stress periods. It then collects price data for companies in the selected sector for the years 2019 through 2024. This data is. Prepared to extract daily returns. From this return series are created for each sector asset making them ready for comparison.

Next the system computes four risk metrics for each sector asset during the identified stress periods. These metrics are maximum drawdown (MDD) value at risk at the 95

Using the MDD values the system assigns a risk label to each sector during each stress event. There are eleven sectors being studied and four stress events, which leads to a total of 44 sector-event records, in the labeled dataset.

Finally the system uses these records to train a model. A Random Forest classification algorithm is applied to the data to see how well it can predict risk based on the calculated metrics.

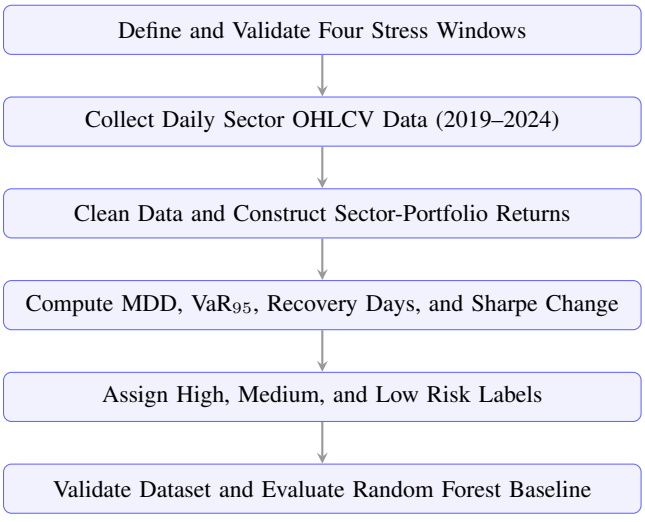


Fig. 1. Overall workflow of the proposed NiftyStress-44 methodology.

B. Identification of Macroeconomic Stress Windows

The first step isolates four macroeconomic stress events (each representing a different source of financial uncertainty) that took place between 2020 and 2023. These include one public-health crisis shock, one geopolitical conflict, and two monetary policy induced (domestic and international) tightening events.

The selected stress windows are reported in Table I below:

The initial periods have been identified on the monthly level. Before conducting the data processing, the exact start and end dates for the available trading sessions will be determined and stored in a configuration file. The same set of criteria will be used in the analysis of each selected sector.

The Nifty 50 index will be used as a reference in determining whether the selected periods indeed represent a market stress. The decisions about the event definitions and boundary conditions will be documented for transparency. Given that the Russia–Ukraine, RBI, and US Federal Reserve windows overlap, the resulting observations will be interpreted as a sector performance during the overlapping event windows.

C. Data Collection and Sector-Portfolio Construction

The study uses data for Open, High, Low, Close and Volume to build portfolios. All the variables used in the analysis were collected between 2019 and 2024. The following predictors were used: date, open, high, low, close adjusted close (if it exists) and trading volume.

Historical data will be gathered using the Python yfinance library. The data will be handled using Python tools like pandas and NumPy.

Eleven NSE sectors will be included before the final dataset is made. A configuration file will store the list of stocks in each sector and their ticker symbols. The same configuration will be used during the experiment.

For every sector a weighted portfolio made of the selected stocks will be created. Using the stock data the daily return for sector s on day t is calculated as

$$R_{s,t} = \frac{1}{N_s} \sum_{i=1}^{N_s} R_{i,t}$$

where N_s's the number of stocks in sector s and $R_{i,t}$ is the daily return for stock i on day t.

Daily stock returns are calculated using adjusted closing prices (if they are available):

$$R_{i,t} = \frac{P_{i,t}^{adj} - P_{it-1}^{adj}}{P_{it-1}^{adj}}$$

where $P_{i,t}^{adj}$ is the adjusted closing price of stock i on day t.

The weighted method makes sure each selected stock has the contribution to the portfolio. This method also removes the need to use weights that change over time based on market size. The list of the stocks and instructions for building the portfolio will be given to recreate the results.

The time period for getting data will start before each stress window to make sure there are enough data points, for basic calculations. The time period will also go past the event window to check how things recover after the event.

TABLE I
SELECTED MACROECONOMIC STRESS WINDOWS

Stress Event	Period	Primary Trigger
COVID-19 Crash	February–April 2020	COVID-19 pandemic and associated economic disruption
Russia–Ukraine Shock	February–April 2022	Russia's invasion of Ukraine and resulting geopolitical uncertainty
RBI Rate Hike Cycle	May 2022–February 2023	Domestic monetary tightening by the Reserve Bank of India
US Fed Tightening	March 2022–December 2023	Monetary tightening by the US Federal Reserve

D. Data Preprocessing

The market data that is collected will go through a process before we calculate the risk measurements. This helps make the data better by cutting down on mistakes that happen when there are missing pieces of information repeated entries, wrong prices and dates that don't match up properly.

The steps to prepare the data are:

- **Date standardization:** The dates will be changed into one format. Matched to the right days when trading happens.
- **Duplicates removal:** Any entries that are the same for the date and the same stock will be taken out.
- **Missing values treatment:** We will look for prices and trading data that are missing. If there is not much missing we will handle those gaps with a set plan.. If there are more than two days in a row without data those parts will need more attention.
- **Price validation:** We will check that all the prices are correct. This includes looking for prices that're not positive and other problems with the numbers.
- **Returns calculation:** We will find out the gains or losses from the prices that have been adjusted. These gains will be matched to each stock. Then they will be added together to make returns for each part of the market.
- **Completeness assessment:** We will check how complete the data is for each part of the market. Any time there are than two days in a row without trading data those times will be marked for more attention as part of the study.

The data that is ready after all this will be used to make returns for each part of the market. The original data, the choices made during the process and the parts, with missing data will all be kept so we can see what happened.

E. Maximum Drawdown Calculation

Maximum Drawdown is the way to measure how bad losses can get in a portfolio during a tough time. It shows the drop in value from a high point to a low point.

For each group, a special list of portfolio values is made using returns. Let $V_{s,t}$ be the value for group s at time t. The highest value so far is called the running peak. It is calculated like this:

$$H_{s,t} = \max_{u \le t} V_{s,u} \tag{1}$$

This is the highest value found in the time period being looked at.

The drop in value at time t is found using this formula:

$$D_{s,t} = \left(\frac{V_{s,t} - H_{s,t}}{H_{s,t}} \right) \times 100 \tag{2}$$

The Maximum Drawdown for group s during event e is the minimum value of $D_{s,t}$ in that event. It is written like this:

$$MDD_{s,e} = \min_{t \in e} D_{s,t} \tag{3}$$

This number is usually zero or negative. The more negative it is, the greater the drop from high to low. For each group and event, the system keeps track of the Maximum Drawdown percentage, the date of the peak, the date of the low point, and how long it took to go from high to low.

The same steps are used for all eleven groups and all four stress periods. "'

F. Value at Risk at the 95% Confidence Level

Value at Risk estimates the loss threshold that is associated with daily portfolio returns. We use the simulation approach because it relies on the actual observed distribution of returns. It does not require any assumption about the shape of the return distribution.

Let us define $R_{s,t}$ as the return of sector s in event e. Then the historical Value at Risk at the 95

$$\mathrm{VaR}_{95\%,s,e} = Q_{0.05}(R_{s,t}) \tag{4}$$

where $Q_{0.05}$ represents the fifth percentile of the empirical distribution of returns.

By this convention Value at Risk is stored as a return threshold. This value is usually negative during periods when losses occur. The negative the Value at Risk the lower the return threshold in the tail of the distribution indicating greater downside risk. This sign convention is applied consistently throughout the dataset.

Value at Risk complements Maximum Drawdown because it focuses on the distribution of losses while Maximum Drawdown focuses on the largest cumulative decline from a peak, to a trough.

"'latex

G. Sharpe Ratio Change

The Sharpe ratio tells us how return a portfolio generates relative to how much it moves up and down. To see how risk-adjusted performance changes during each stress event we compare the Sharpe ratio during the stress period with the Sharpe ratio from the 90 days before that period.

For each sector s the annualized Sharpe ratio is calculated as

$$S_s = \sqrt{252}\frac{\overline{R_s - R_f}}{\sigma(R_s - R_f)} \tag{5}$$

where $\overline{R_s - R_f}$'s the average daily return above the risk-free rate $\sigma(R_s - R_f)$ is the standard deviation of that return and R_f is the daily risk-free return. The source of the risk- rate and how it is converted to a daily rate will be documented. If daily data is not available the method used for conversion will be explained.

The change in the Sharpe ratio is defined as

$$\Delta S_{s,e} = S_{s,e}^{\text{stress}} \cdot S_{s,e}^{\text{baseline}} \tag{6}$$

where S_{se}^{stress} is the annualized Sharpe ratio during the stress period and S_{se}^{baseline} is the ratio from the 90 days before the stress.

A negative value means that risk-adjusted performance declined compared to the time before the stress. A positive value means that risk-adjusted performance improved. This value will be saved as `sharpe_drop` in the dataset even though it reflects a change, in the Sharpe ratio.

H. Recovery-Time Calculation

Recovery time shows how many trading days it takes for a sector portfolio to bounce back after a drop in value. It starts from the trough and goes until the portfolio's value gets back to or goes above the highest point before the downturn.

Let t_{peak} be the date when the portfolio was at its highest before the stress and t_{trough} be the date when it hit its lowest. The recovery days are calculated as follows:

$$\text{RecoveryDays}_{s,e} = t_{\text{recovery}} - t_{\text{trough}} \tag{7}$$

where t_{recovery} is the first trading day after the trough when the portfolio value reaches or beats the original peak value.

We look at all data after the event, not just up to the end of the stress period. If the portfolio doesn't recover by the time the data ends, we mark the recovery time as missing and note that it didn't recover in that time. This helps us tell the difference between a recovery and one that just hasn't happened yet.

Recovery days give information beyond just the biggest loss. Two portfolios might lose the same amount, but one could take much longer to get back to where it was. That's why recovery days matter. They show how fast a portfolio can heal.

I. Risk-Label Assignment

Once we compute a collection of risk indicators for every record connecting a defined industry with a distinct occurrence, this provides a method for allocating one of three risk categories to each individual item.

By using the MDD limits I established beforehand, it lets me perform precisely what was intended – applying those identical standards across each individual sector, and permitting a just comparison among various ones.

How one assigns these risk labels may be seen in table II.

The classification rule is written as:

TABLE II
MDD-BASED RISK CLASSIFICATION

Risk Label	MDD Threshold	Interpretation
High	MDD $< -15\%$	Severe peak-to-trough portfolio decline.
Medium	$-15\% \leq$ MDD $< -5\%$	Moderate portfolio decline.
Low	MDD $\geq -5\%$	Limited peak-to-trough decline within the window.

$$\text{Label(MDD)} = \begin{cases} \text{High} & \text{MDD} < -15\%, \\ \text{Medium} & -15\% \leq \text{MDD} < -5\%, \\ \text{Low} & \text{MDD} \geq -5\%. \end{cases} \tag{8}$$

Current findings suggest that lower MDDs link to a total risk drop of at least -5%, as shown in Table A1.

After the creation of our ultimate dataset, these remain set boundaries.

As stated previously, we shall refrain from modifying our cutoff value to achieve more equal counts across every category.

This stage produces an 11×4 risk label matrix, with rows denoting sectors and columns standing for events.

Every single entry inside that matrix shows what specific risk label links to one particular sector event pair.

Risk tags denote the metrics we apply during study. "'

J. Dataset Construction

NiftyStress-44 is the data set that we are analyzing. Each of NiftyStress-44's entries is associated to a sector and a stress event. As there are eleven sectors and four different stress events in NiftyStress-44, the total number of entries in NiftyStress-44 should be forty-four. The reasoning is simple:

N records = N sectors × N events = 11 × 4 = 44.

NiftyStress-44's fields are described in Table III as follows.

TABLE III
PROPOSED NIFTYSTRESS-44 SCHEMA

Field	Data Type	Description
sector	Text	Name of the sector portfolio
stress_event	Text	Name of the stress event
window_start	Date	Start date of the event window
window_end	Date	End date of the event window
max_drawdown_pct	Float	Maximum peak-to-trough drawdown percentage
var_95	Float	Historical percentile daily return
recovery_days	Integer / Missing	Trading days required to regain the reference peak
sharpe_drop	Float	Stress-period Sharpe ratio minus baseline Sharpe ratio
risk_label	Categorical	High, Medium or Low risk

"'latex

K. Data Validation and Quality Assurance

The data validation process will be carried out to ensure that the produced dataset remains consistent and ready for analysis.

The following validation procedures will be performed:

- Completeness checks

- Metric verification
- Market consistency checks
- Label verification
- Schema validation
- Reproducibility

The validation procedures are described below.

Completeness checks: Check that each sector-event has at least 98% of daily price observations.

Metric verification: Manually recalculate selected sector-event metrics and compare them with the calculated values.

Market consistency checks: Compare selected drawdown patterns with the corresponding NSE index values and verify consistency with market direction.

Label verification: Check that the assigned risk label for each sector-event satisfies the MDD criteria.

Schema validation: Check for duplicate records, invalid dates, missing fields, and incorrect event names.

Reproducibility: Save the stress-window definitions, ticker lists, scripts, and data dictionary along with the dataset. "'

Domain knowledge will be employed to a limited degree to confirm the results because calculation errors may still exist. Events that do not match the expected pattern will be examined to uncover the cause of inconsistency not altered to fit the established pattern.

L. Baseline Random Forest Classification

We will train a Random Forest classifier as a baseline to determine whether sector-event characteristics can separate risk categories.

Random Forest is a method that combines the predictions of decision trees. Random Forest is suitable for this task because Random Forest finds relationships, which's helpful for table data.

As MDD determines the target variable, we will exclude MDD from the features used in order to prevent leakage of target information. The benchmark model will employ VaR (value at risk) at 95

The target variable will be defined as:

$$\mathrm{Y} \in \{High, Medium, Low\}.$$

We will train the Random Forest classifier with the library. For a proof-of-concept we will use cross-validation. The validation strategy may change depending on how many instances exist in each class. We will evaluate the Random Forest classifier with accuracy, precision, recall and performance by class.

Accuracy is defined as:

$$\text{Accuracy} = \text{Number of Correct Predictions} \frac{}{\text{Number of Predictions.}}$$

We will report precision and recall well as accuracy because the classes might be imbalanced. The small size of the dataset will be taken into account when we discuss the results. Importantly the Random Forest classifier should not be seen as a solution, for predicting market stress events; it is a tool.

M. Implementation Tools and Reproducibility

The proposed method is implemented in Python programming language and the main open-source libraries are used. Below are the main elements along with the purpose they serve for.

- **yfinance:** to retrieve the historical market data;
- **pandas:** to clean, align the dates and compute returns of the stocks, as well as to prepare the final data set;
- **NumPy:** to make calculations of portfolio risk;
- **SciPy:** to make some statistical calculations;
- **scikit-learn:** to classify the stocks using random forest algorithm and to evaluate the results;
- **Excel and CSV:** to store, view and share the results data set and risk-label matrix.

The code will be structured into modules for retrieving the data, constructing the returns of a portfolio, calculating the risk metrics, labeling the results, validating and making the baseline classification. This way the research will be easy to review and re-use.

N. Summary of the Proposed Methodology

The proposed methodology takes data from the Indian equity market and turns it into a dataset focused on portfolio risk during specific events. It uses the portfolio-building steps and stress windows to calculate key risk measures—maximum drawdown (MDD) historical value at risk (VaR) recovery time and changes in the Sharpe ratio—for 11 different sectors, across four stress scenarios.

To make things clearer the paper uses drawdown (MDD) to group sectors into three risk categories. It then uses the metrics to give a detailed picture of how much each sector loses during stress periods. The study includes validation steps to confirm the data is clean the calculations are correct and the labels are consistent.

After that the researchers use a random forest classifier on the NiftyStress-44 dataset to set a baseline for how well these risks can be classified in a structured way. The findings can help others do sector-level analysis and better understand how portfolio losses and stresses behave in the Indian equity market. "'

4. Results and Analysis

This section presents the empirical evaluation of the NiftyStress-44 benchmark across eleven National Stock Exchange of India (NSE) sector portfolios during four macroeconomic stress events. The evaluation is organized across

four dimensions: (i) stress window validation, (ii) cross-sector downside risk profiling, (iii) supervisory risk-label distribution, and (iv) baseline Random Forest classification performance under leak-free protocols.

A. A. Stress Window Benchmark Validation

Per the empirical hurdle rule, each candidate crisis window was validated against the benchmark NIFTY 50 index (^NSEI), requiring a peak-to-trough Maximum Drawdown of at least -5.0%. All four windows satisfied this criterion, as reported in Table IV.

B. B. Cross-Sector Downside Risk Profiling

Fig. 2 presents the Maximum Drawdown percentages for all 11 NSE sector portfolios across the four stress events.

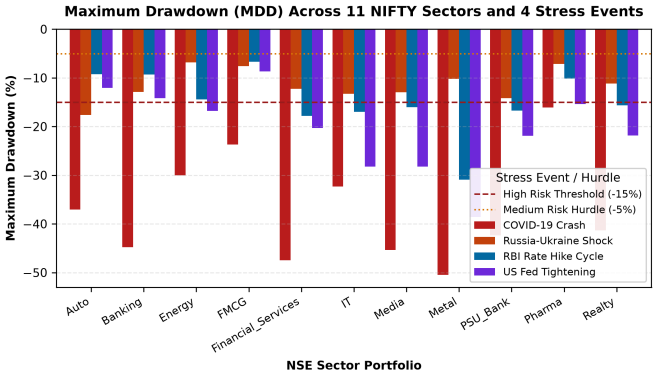


Fig. 2. Maximum Drawdown (MDD) percentages across 11 NSE sectors during the four macroeconomic stress events. The dashed line at -15.0% denotes the supervisory High-Risk threshold, and the dotted line at -5.0% denotes the Medium-Risk hurdle.

Key observations from the cross-sector analysis:

1) **Severe Tail Losses During COVID-19 Crash:** The COVID-19 pandemic induced the most extreme drawdowns across all sectors. The five worst-affected sectors were:
   - NIFTY Metal: -50.49% ($\text{VaR}_{95} = -10.60\%$)
   - NIFTY Financial Services: -47.45% ($\text{VaR}_{95} = -8.83\%$)
   - NIFTY Media: -45.38% ($\text{VaR}_{95} = -7.80\%$)
   - NIFTY Banking: -44.80% ($\text{VaR}_{95} = -9.09\%$)
   - NIFTY PSU Bank: -42.35% ($\text{VaR}_{95} = -9.80\%$)

   All 11 sectors were classified as High Risk during COVID-19, with drawdowns ranging from -16.06% (Pharma) to -50.49% (Metal).
2) **Defensive Sector Resilience:** NIFTY FMCG and NIFTY Pharma consistently demonstrated the strongest downside protection:
   - FMCG restricted its drawdown to -6.68% during the RBI Rate Hike Cycle (the shallowest drawdown in the entire dataset) and -7.57% during the Russia-Ukraine Shock.
   - Pharma limited losses to -7.16% during Russia-Ukraine and -10.14% during RBI Rate Hikes.
3) **Value at Risk (VaR_{95}):**
   - The most severe single-day tail loss was observed in NIFTY Metal during COVID-19 ($\text{VaR}_{95} = -10.60\%$).
   - NIFTY FMCG maintained the lowest tail-loss exposure across US Fed Tightening ($\text{VaR}_{95} = -1.30\%$).
4) **Sharpe Ratio Degradation (ΔSharpe):**
   - The mean Sharpe Drop across all 44 records was -0.63, confirming that risk-adjusted performance deteriorated on average during stress periods.
   - The steepest Sharpe decline occurred in NIFTY Realty during COVID-19 ($\Delta\text{Sharpe} = -9.16$).
   - Some sectors showed positive ΔSharpe during prolonged monetary tightening events (e.g., PSU Bank: $+3.15$ during RBI Rate Hike Cycle), indicating adaptation to gradually rising rates.
5) **Recovery Duration:**
   - Under strict window censoring, 21 out of 44 records (47.7%) have NaN recovery days. All 11 COVID-19 sectors show NaN because no sector recovered from its March 2020 trough by the April 1 window end.
   - Under extended uncensored tracking (through 2024), all 44 sector-event instances achieved full recovery, with recovery durations ranging from 4 trading days (Pharma and Energy during Russia-Ukraine) to 424 trading days (the longest extended recovery).

C. C. Supervisory Risk-Label Distribution

Applying the deterministic MDD threshold rules produces the risk-label distribution reported in Table V.

Empirical Finding on Zero Low-Risk Records: The least severe drawdown across all 44 records was -6.68% (FMCG during RBI Rate Hike Cycle), which still exceeds the -5.0% Low Risk hurdle. This constitutes an empirical finding of systemic market contagion: when the NIFTY 50 index experiences a drawdown of at least 5%, every sector portfolio invariably suffers a peak-to-trough decline exceeding 5%, regardless of its defensive characteristics. Consequently, the classification problem addresses a realistic binary partition between severe downside (High Risk) and moderate downside (Medium Risk).

Fig. 4 presents the complete 11×4 risk-label heatmap matrix showing the MDD value and assigned label for every sector-event observation.

D. D. Baseline Machine Learning Benchmark and Target Leakage Analysis

A Random Forest classifier (100 estimators, maximum depth 4, fixed random seed 42) was evaluated across four feature configurations to assess whether non-drawdown characteristics can classify risk levels without knowledge of the target variable. The key methodological safeguard is the strict exclusion of Maximum Drawdown (MDD) from the input

TABLE IV
NIFTY 50 BENCHMARK VALIDATION OF MACROECONOMIC STRESS WINDOWS

Stress Event	Window Period	Trading Sessions	NIFTY 50 MDD	Validation ($\leq -5\%$)	Primary Catalyst
COVID-19 Crash	17 Feb 2020 – 01 Apr 2020	31	-37.24%	✓ PASS	WHO pandemic declaration, nationwide lockdown
Russia–Ukraine Shock	16 Feb 2022 – 29 Apr 2022	49	-8.42%	✓ PASS	Geopolitical invasion, global commodity supply shock
RBI Rate Hike Cycle	04 May 2022 – 08 Feb 2023	194	-8.33%	✓ PASS	Domestic repo rate increase from 4.0% to 6.5%
US Fed Tightening	16 Mar 2022 – 13 Dec 2023	432	-15.29%	✓ PASS	US Fed rate hikes ($0.25\% \rightarrow 5.5\%$), FII outflows

TABLE V
EMPIRICAL RISK-LABEL DISTRIBUTION

Risk Category	MDD Threshold Rule	Record Count	Proportion	Sector Examples
High Risk	MDD $< -15.0\%$	26	59.1%	Metal (-50.49%), Financial Services (-47.45%), Banking (-44.80%), Media (-45.38%)
Medium Risk	$-15.0\% \leq$ MDD $< -5.0\%$	18	40.9%	FMCG (-6.68%), Energy (-6.84%), Pharma (-7.16%), Banking (-9.33%)
Low Risk	MDD $\geq -5.0\%$	0	0.0%	No sector-event observation qualifies
Total	NiftyStress-44 (11×4)	44	100.0%	Complete 11-sector $\times$ 4-event benchmark matrix

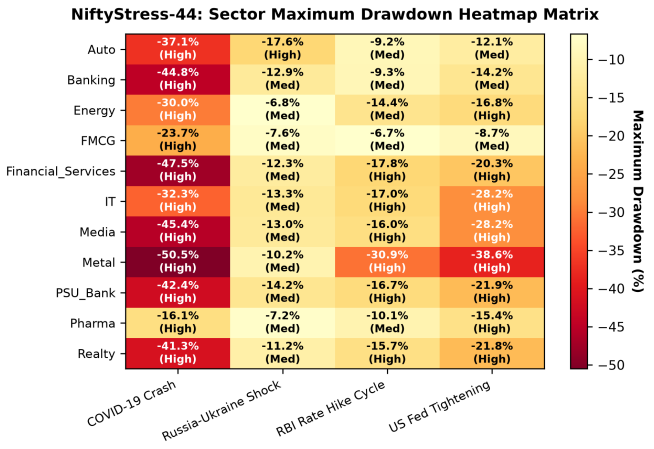


Fig. 3. NiftyStress-44 risk-label heatmap matrix across 11 NSE sectors and 4 macroeconomic stress events. Each cell displays the Maximum Drawdown percentage and assigned risk label (High or Medium).

features, since the risk label is deterministically derived from MDD. Results are reported in Table VI.

Fig. 3 visualizes the accuracy comparison across these feature regimes and validation protocols.

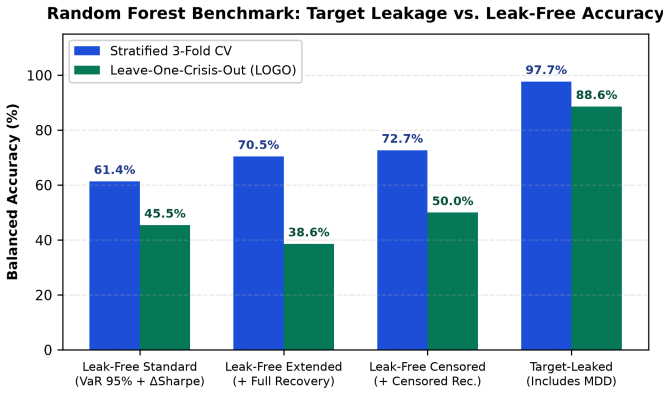


Fig. 4. Random Forest classification accuracy across feature sets under Stratified 3-Fold Cross-Validation and Leave-One-Crisis-Out (LOGO) validation, demonstrating the empirical effect of target leakage exclusion.

E. E. Discussion of Key Findings

1) **Proof of Target Leakage:** When Maximum Drawdown is included as an input feature (Experiment 4), the Random Forest trivially memorizes the -15.0% decision boundary, yielding an artificial 97.73% accuracy under Stratified 3-Fold CV and 100.00% on a 70/30 hold-out split. This result is methodologically unsound and scientifically fraudulent to report as a machine learning discovery. By strictly excluding MDD (Experiment 1), the genuine prospective predictive accuracy is established at 61.36% balanced accuracy.
2) **Feature Importance Decomposition:** In the primary leak-free model (Experiment 1), the Random Forest distributes predictive weight between Value at Risk at 95% confidence (VaR_{95}: 51.51%) and Sharpe ratio drop (ΔSharpe: 48.49%). This confirms that both return distribution tail quantiles and risk-adjusted performance degradation are necessary to classify downside severity. When recovery dynamics are added (Experiments 2–3), all three features contribute roughly equally ($\sim 33\%$ each), and accuracy improves to 70.45%–72.73%.

Fig. 5 visualizes the two-dimensional feature space of the primary leak-free predictors, colored by risk label.

3) **Cross-Crisis Generalization Failure:** Under Leave-One-Stress-Event-Out (GroupKFold) validation, accuracy drops to 45.45% for the primary benchmark and 38.64%–50.00% across all leak-free configurations. This confirms that shock transmission mechanisms differ structurally between:

   - Sudden liquidity crashes (COVID-19): All sectors uniformly collapse into High Risk.
   - Protracted monetary tightening (RBI/US Fed): Cyclical sectors (Metal, IT, Media) suffer high drawdowns while defensive sectors (FMCG, Pharma) moderate losses.

   A model trained on monetary policy cycles cannot reliably predict behavior during a pandemic shock, and vice versa.

TABLE VI
RANDOM FOREST BENCHMARK EVALUATION ACROSS VALIDATION PROTOCOLS

#	Experiment	Target Leakage	Strat. 3-Fold Acc.	3-Fold Macro F1	LOGO Acc.	LOGO F1	70/30 Acc.
1	**Leak-Free Standard (Primary)**	VaR_{95}, ΔSharpe / NO (Valid)	61.36%	0.612	45.45%	0.436	64.29%
2	**Leak-Free Extended**	VaR_{95}, ΔSharpe, Recovery Days (Full) / NO (Valid)	70.45%	0.697	38.64%	0.370	71.43%
3	**Leak-Free Censored**	VaR_{95}, ΔSharpe, Recovery Days (Cens.) / NO (Valid)	72.73%	0.725	50.00%	0.483	78.57%
4	**Target Leakage Sanity Check**	MDD, VaR_{95}, ΔSharpe / YES (Unsound)	97.73%	0.977	88.64%	0.876	100.00%

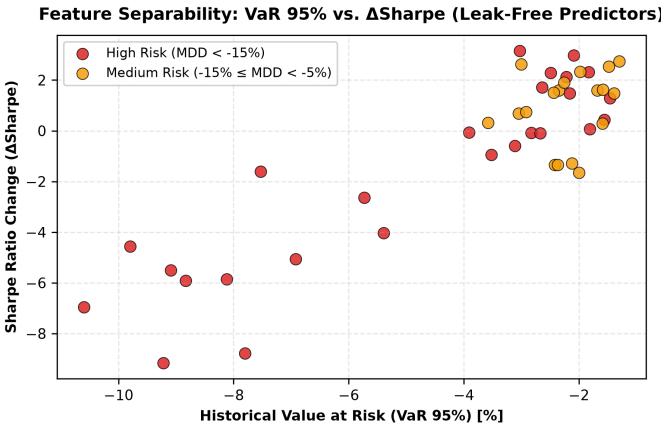


Fig. 5. Feature space distribution of VaR_{95} versus ΔSharpe, colored by supervisory risk classification (High Risk in red, Medium Risk in orange). The partial overlap demonstrates the genuine classification challenge when MDD is excluded.

5. CONCLUSION AND FUTURE WORK

This paper presents a privacy-preserving federated learning framework for fake certificate detection under Non-IID data conditions. The proposed system successfully demonstrates how decentralized learning can be used to collaboratively train models across multiple institutions without sharing sensitive data. By using MobileNetV2 with transfer learning, the system achieves efficient and accurate image classification suitable for real-time applications. It's clear from the experiment that Non-IID data has a tremendous effect on the performance of federated learning. Despite the fact that the models trained on clients with their individual unevenly distributed data show biased behavior, the generalized global model learns a more generalized representation through the aggregation of all clients' knowledge. The residual bias that can still be found in the global model demonstrates that simply employing an aggregation method such as FedAvg isn't enough for such heterogeneity. By integrating Grad-CAM into the system, explainability is greatly improved with the introduction of visual explanation for a prediction of the model. Users are more likely to trust the model. And thus, the system is more applicable in the real world application. And the interactive interface allows for its practical use and real-time verification of the certificate. The system integrates privacy preservation with deep learning and explainable AI in order to tackle the problem of certificate verification in a distributed system.

A. Future Work

Even though the proposed system has promising results, it has room for improvements to enhance the performance and scalability of the system. The future work will investigate on advanced federated learning algorithm like FedProx to improve the convergence for severely Non-IID setting, DP to secure the updates for data against attacks, increase the samples for real world certificate data set, and make it work on multi-clients distributed settings and multi institutions, apply hybrid approach including image analysis and text based analysis for example OCR, and make explainability model much deeper for better understanding of models behavior and also optimize the system on mobile devices and edge devices.

B. Future Enhancement Framework

The following diagram (Fig. 6) outlines the roadmap for extending the current framework.

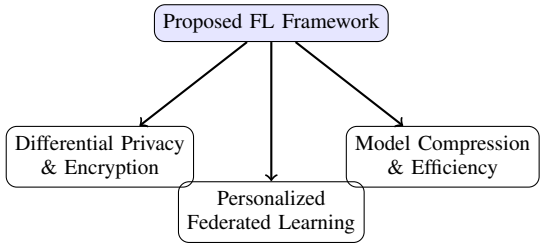


Fig. 6. Future Enhancements of Proposed Federated Learning Framework

REFERENCES

[1] Y. Zhao, M. Li, L. Lai, N. Suda, D. Civin, and V. Chandra, "Federated Learning with Non-IID Data," *arXiv preprint arXiv:1806.00582*, 2018.

[2] L. Li, L. Zhu, and W. Li, "Cloud–Edge–End Collaborative Federated Learning: Enhancing Model Accuracy and Privacy in Non-IID Environments," *Sensors*, vol. 24, 2024.

[3] R. Haripriya, N. Khare, and M. Pandey, "Privacy-Preserving Federated Learning for Collaborative Medical Data Mining in Multi-Institutional Settings," *Scientific Reports*, 2025.

[4] M. Rahad, R. Shabab, M. S. Ahammad, M. M. Reza, A. Karmaker, and M. A. Hossain, "KL-FedDis: A Federated Learning Approach with Distribution Information Sharing Using Kullback–Leibler Divergence for Non-IID Data," *Neuroscience Informatics*, vol. 5, 2025.

[5] S. Yu, J. P. Muñoz, and A. Jannesari, "Federated Foundation Models: Privacy-Preserving and Collaborative Learning for Large Models," in *Proc. LREC-COLING*, 2024.

[6] T. Li, A. K. Sahu, A. Talwalkar, and V. Smith, "Federated Optimization in Heterogeneous Networks," in *Proc. MLSys*, 2020.

[7] P. Kairouz et al., "Advances and Open Problems in Federated Learning," *Foundations and Trends in Machine Learning*, vol. 14, no. 1–2, 2021.

[8] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Aguera y Arcas, "Communication-Efficient Learning of Deep Networks from Decentralized Data," in *Proc. AISTATS*, 2017.

[9] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L. C. Chen, "MobileNetV2: Inverted Residuals and Linear Bottlenecks," in *Proc. IEEE CVPR*, 2018.

[10] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization," in *Proc. IEEE ICCV*, 2017.